"Virtual Memory Page Replacement Visualizer"

A Hackathon Project Report submitted in partial fulfillment of the grade for the course

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25SC2104E)

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Declaration

The Project Report entitled **"Virtual Memory Page Replacement Visualizer"** is a record of bonafide work of **SAI SURENDRANATH PENDYALA – 2520030157, ADA ROHAN – 2520030195, MEDIKONDA RAM RATAN – 2520030291** submitted in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L (Deemed to be) University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the project report entitled "Virtual Memory Page Replacement Visualizer" is a bonafide work done and submitted by Sai Surendranath Pendyala (2520030157), Ada Rohan (2520030195) and Medikonda Ram Ratan (2520030291) in partial fulfillment of the requirements of the course Operating Systems and Systems Programming (25SC2104E) in the Department of Computer Science and Engineering, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Guide

Signature of the Course Coordinator

Signature of the HOD

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1. ABSTRACT

The Virtual Memory Page Replacement Visualizer is a web-based application designed to demonstrate and compare three page replacement algorithms used in operating systems: First In, First Out (FIFO), Least Recently Used (LRU), and Optimal Page Replacement. When physical memory becomes full and a new page needs to be loaded, the operating system must select a page to replace. The proposed system simulates this process using a user-defined page reference string and a specified number of memory frames. It visually displays the contents of memory frames after each page request and identifies page hits and page faults.

The system calculates the total number of page faults generated by each algorithm and presents a comparative analysis using tables and graphical representations. This helps users understand how different page replacement strategies operate and how their performance varies with different input sequences. The application is developed using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface for learning operating system memory management concepts. The proposed visualizer serves as an educational tool for students to understand virtual memory, page

replacement techniques, and the importance of reducing page faults to improve memory management efficiency.

2. INTRODUCTION

Virtual memory is an important concept in operating systems that allows programs to execute efficiently by using secondary storage as an extension of physical memory. When a required page is not available in RAM, a page fault occurs, and the operating system must load the required page into memory. If all memory frames are occupied, a page replacement algorithm is used to select a page for removal. FIFO (First In, First Out), LRU (Least Recently Used), and Optimal Page Replacement are three commonly studied algorithms for this purpose.

Understanding their working process and comparing their performance can be difficult through theoretical explanations alone. The Virtual Memory Page Replacement Visualizer provides an interactive graphical representation of these algorithms, showing memory frame changes, page hits, page faults, and total page fault counts. It helps students understand memory management concepts and analyze the performance of different page replacement strategies using the same input sequence.

3. OBJECTIVES

The main objective of the Virtual Memory Page Replacement Visualizer is to develop a user-friendly web application that demonstrates the working of FIFO, LRU, and Optimal Page Replacement algorithms. The key objectives are:

- To allow users to enter a page reference string and specify the number of memory frames.
- To visually display the memory frame contents after each page request.
- To clearly identify page hits and page faults during the simulation.
- To calculate the total number of page faults generated by each algorithm.
- To compare the performance of the algorithms using tables and graphical representations.
- To improve students' understanding of virtual memory, page replacement techniques, and operating system memory management through an interactive and easy-to-understand learning experience.

4. PROBLEM STATEMENT

Understanding page replacement algorithms in operating systems can be challenging for students because the process of managing memory frames and replacing pages is difficult to visualize through theoretical explanations alone. When physical memory becomes full, the operating system must select a page to remove before loading a new page. Different algorithms, such as FIFO, LRU, and Optimal Page Replacement, make these decisions using different strategies, resulting in different numbers of page faults.

Therefore, there is a need for an interactive visualizer that demonstrates the step-by-step working of these algorithms, displays page hits and page faults, and compares their performance using the same page reference string and number of memory frames. The proposed system aims to simplify the learning process and help users understand the effectiveness of different page replacement techniques.

5. EXISTING SOLUTION

Traditional methods of learning page replacement algorithms mainly rely on textbooks, classroom explanations, and static examples. These methods explain the working principles of FIFO, LRU, and Optimal Page Replacement but may not provide an interactive representation of memory frame changes for different page reference strings.

Some existing simulators demonstrate individual algorithms or require users to calculate page faults manually. Comparing multiple algorithms can therefore be time-consuming and difficult for beginners. These limitations highlight the need for a user-friendly visual tool that demonstrates page replacement step by step and compares the performance of all three algorithms using the same input.

6. PROPOSED SOLUTION

The proposed Virtual Memory Page Replacement Visualizer is an interactive web application designed to simplify the understanding of page replacement algorithms. The system allows users to enter a page reference string and specify the number of memory frames. It simulates FIFO, LRU, and Optimal Page Replacement algorithms and displays the changes in memory frames step by step.

Page hits and page faults are highlighted visually to help users understand how each algorithm handles page requests. The application calculates the total page faults for each algorithm and presents a comparative analysis using tables and graphical representations. By providing a simple interface and real-time visual feedback during simulation, the proposed system helps students understand memory management concepts and evaluate the performance of different page replacement strategies.

7. TECHNOLOGIES USED

The Virtual Memory Page Replacement Visualizer is developed using HTML, CSS, and JavaScript. These technologies work together to provide an interactive and efficient platform for understanding virtual memory and page replacement techniques.

- **HTML:** used to structure the web application, including input fields, buttons, memory frames, and result sections.
- **CSS:** used to design an attractive, responsive, and user-friendly interface with visual indicators for page hits and page faults.
- **JavaScript:** implements the FIFO, LRU, and Optimal Page Replacement algorithms, processes the user-provided page reference string, updates memory frame contents, and calculates the total number of page faults.

Tables and graphical representations are used to compare the performance of the three algorithms.

8. SYSTEM DESIGN

The system design of the Virtual Memory Page Replacement Visualizer consists of three main components:

- **User Interface:** allows users to enter a page reference string and specify the number of memory frames.
- **Page Replacement Algorithm Module:** processes the input using FIFO, LRU, and Optimal Page Replacement techniques and determines which pages should be loaded into or removed from memory.
- **Results Visualization Module:** displays the step-by-step simulation, calculates the total page faults for each algorithm, and presents a comparative analysis using tables and charts.

During the simulation, the system records memory frame changes and identifies page hits and page faults. These components work together to provide an interactive and user-friendly platform for understanding and comparing page replacement algorithms.

9. FLOWCHART

The flowchart below shows the overall flow of the system, from taking the user input and validating it, to running each algorithm on the same page reference string and frame count, and finally displaying the results and comparative analysis.

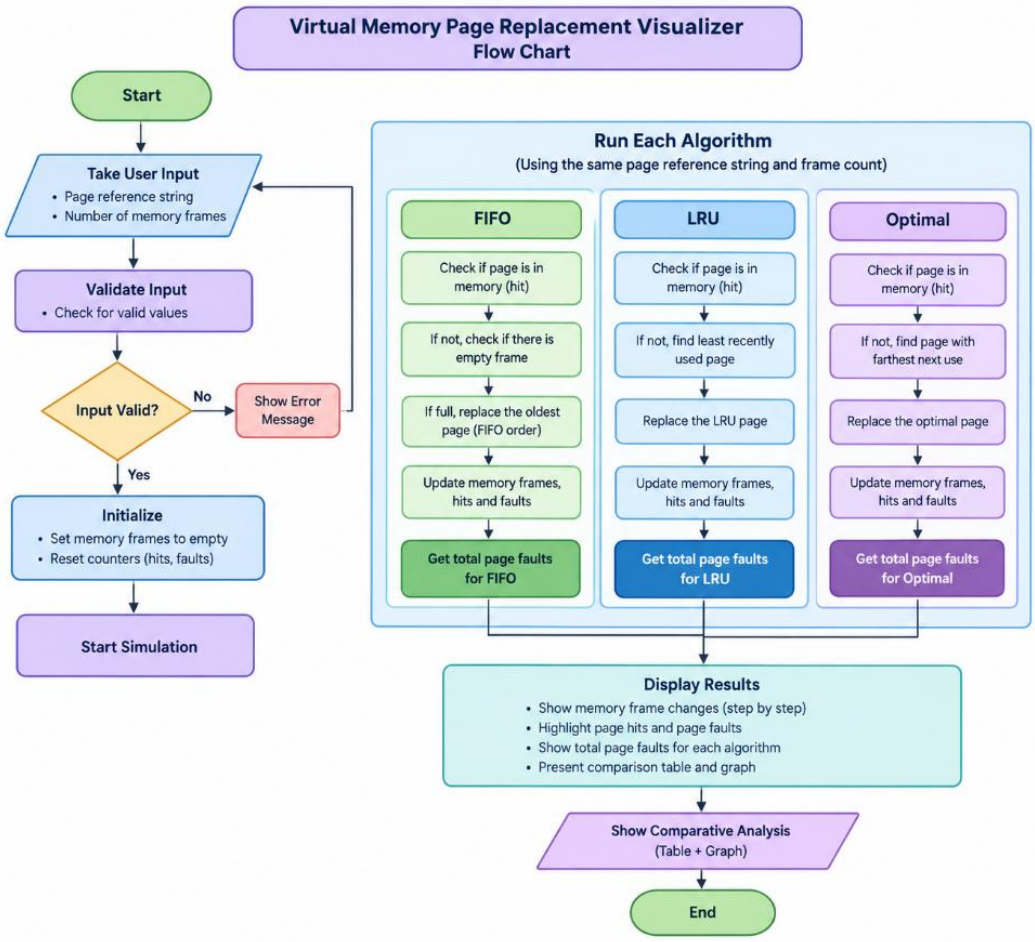


Figure 9.1: Flowchart of the Virtual Memory Page Replacement Visualizer

10. IMPLEMENTATION APPROACH

The implementation of the Virtual Memory Page Replacement Visualizer begins with designing a user-friendly web interface using HTML and CSS. Users can enter a page reference string and specify the number of memory frames required for the simulation. JavaScript is used to implement the FIFO, LRU, and Optimal Page Replacement algorithms.

The system processes each page request, checks whether the page is already present in memory, and identifies page hits and page faults. When all memory frames are occupied, the appropriate algorithm selects a page for replacement. The application updates the memory frame contents after each request and calculates the total number of page faults for each algorithm.

Finally, the results are displayed using step-by-step visualizations, comparison tables, and graphical representations. The system is tested with different input sequences and frame counts to verify the correctness of the algorithms and ensure reliable performance.

11. WEB APPLICATION SCREENSHOT

The screenshot below shows the home page of the web application, with the simulation setup section and the summary cards for the FIFO, LRU, and Optimal algorithms.

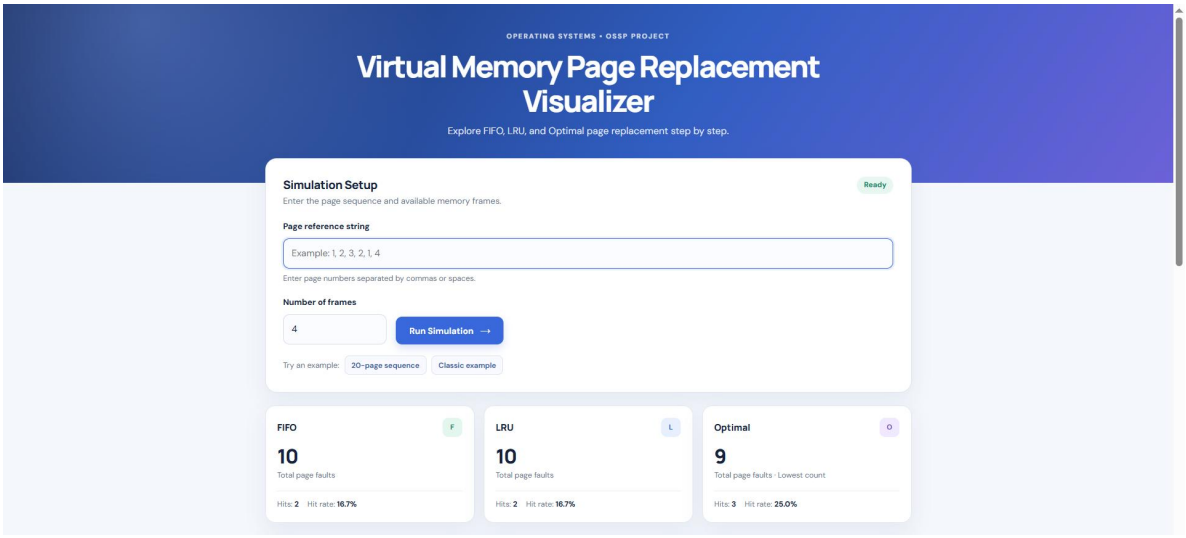


Figure 11.1: Web application screenshot

12. INPUT SCREENSHOTS

The user enters the page reference string and the number of memory frames in the Simulation Setup section and clicks the Run Simulation button. In the sample run, the page reference string is 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 with 3 memory frames.

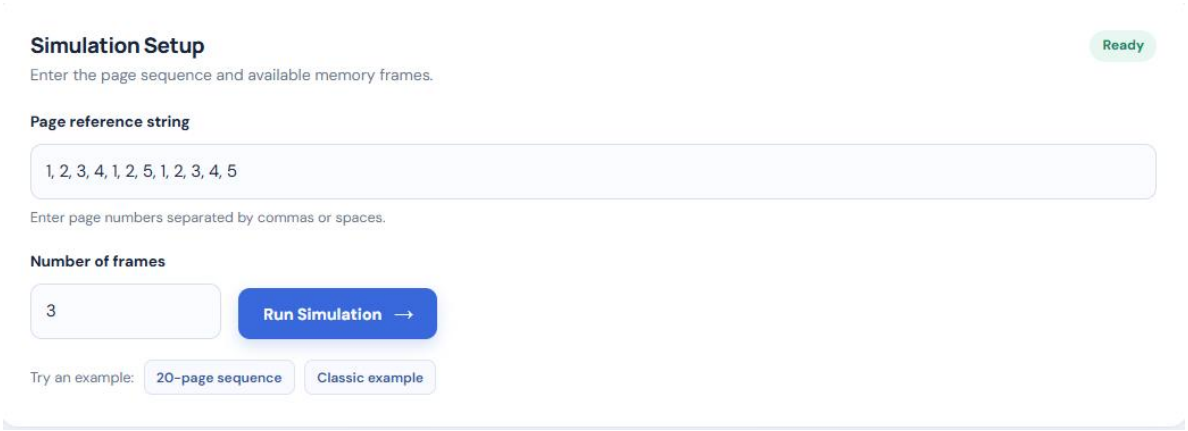


Figure 12.1: Input screenshot (Simulation Setup)

13. OUTPUT SCREENSHOTS

After the simulation is run, the application displays the total page faults, hits, and hit ratio for each algorithm along with a bar chart that compares the page faults generated by FIFO, LRU, and Optimal. For the sample input, Optimal produced the fewest page faults.

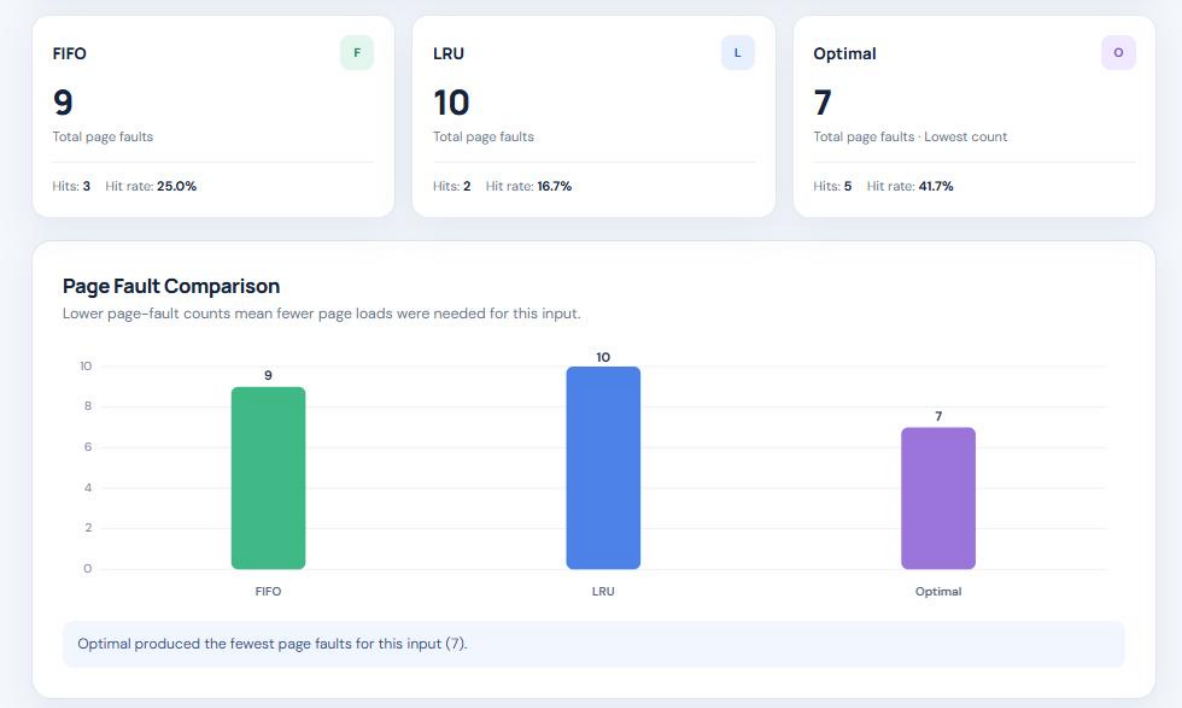


Figure 13.1: Output screenshot (results and page fault comparison)

The comparison of the three algorithms for the sample input is summarized below.

Algorithm	Page Faults	Hits	Hit Ratio
FIFO	9	3	25.0%
LRU	10	2	16.7%
Optimal	7	5	41.7%

Table 13.1: Comparison of page replacement algorithms (12 requests, 3 frames)

14. STEP BY STEP SIMULATION

The step-by-step simulation table shows one column for each requested page. The contents of every memory frame are displayed after each request, with page hits highlighted in green and page faults highlighted in rose, along with a running fault count.

14.1 FIFO Simulation

FIFO (First In, First Out) replaces the page that has been in memory the longest.



Step-by-Step Simulation

Each column represents one requested page. Green indicates a hit; rose indicates a fault.

FIFO | LRU | Optimal

FIFO (First In, First Out) replaces the page that has been in memory the longest.

Page request	1 #1	2 #2	3 #3	4 #4	1 #5	2 #6	5 #7	1 #8	2 #9	3 #10	4 #11	5 #12
Frame 1	1	1	1	4	4	4	5	5	5	5	5	5
Frame 2	—	2	2	2	1	1	1	1	1	3	3	3
Frame 3	—	—	3	3	3	2	2	2	2	2	4	4
Status	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	HIT	HIT	FAULT	FAULT	HIT
Fault count	1	2	3	4	5	6	7	7	7	8	9	9

Page hit Page fault Empty frame

Figure 14.1: Step-by-step simulation of FIFO

14.2 LRU Simulation

LRU (Least Recently Used) replaces the page that has not been accessed for the longest time in the past.



Figure 14.2: Step-by-step simulation of LRU

14.3 Optimal Simulation

Optimal replaces the page whose next use is farthest in the future, or a page that will never be used again.



Figure 14.3: Step-by-step simulation of Optimal Page Replacement

15. IMPACT

The Virtual Memory Page Replacement Visualizer improves students' understanding of virtual memory and page replacement algorithms through interactive visual demonstrations. It helps learners understand how FIFO, LRU, and Optimal algorithms manage memory frames and handle page faults. By comparing the number of page faults generated by each algorithm, students can easily identify differences in their performance.

The visual representation of page hits, page faults, and memory frame changes makes complex operating system concepts easier to learn and remember. This application also supports practical learning, encourages algorithmic thinking, and helps students develop a better understanding of efficient memory management in operating systems.

16. CONCLUSION

The Virtual Memory Page Replacement Visualizer provides an interactive platform for understanding and comparing FIFO, LRU, and Optimal Page Replacement algorithms. It demonstrates memory frame changes, identifies page hits and page faults, and calculates the total page faults generated by each algorithm. The comparative analysis helps users understand how different page replacement strategies perform for the same page reference string.

Developed using HTML, CSS, and JavaScript, the application simplifies operating system memory management concepts through clear visual representations. Overall, the project serves as an effective educational tool for students to learn and understand virtual memory and page replacement techniques.

17. FUTURE SCOPE

The Virtual Memory Page Replacement Visualizer can be enhanced by introducing step-by-step playback controls, animated page replacement, and additional page replacement algorithms such as LFU (Least Frequently Used). Future improvements may include:

- Downloadable simulation reports.
- Interactive quizzes.
- Detailed performance statistics.
- Support for larger page reference strings.
- Customizable visualization themes.
- Mobile-friendly features.

These enhancements can make the system more interactive, accessible, and useful for students learning advanced operating system concepts.